ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

Project Proposal Report

SWE 4304: SPL-I

Project Title- School Admission System

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Project Overview

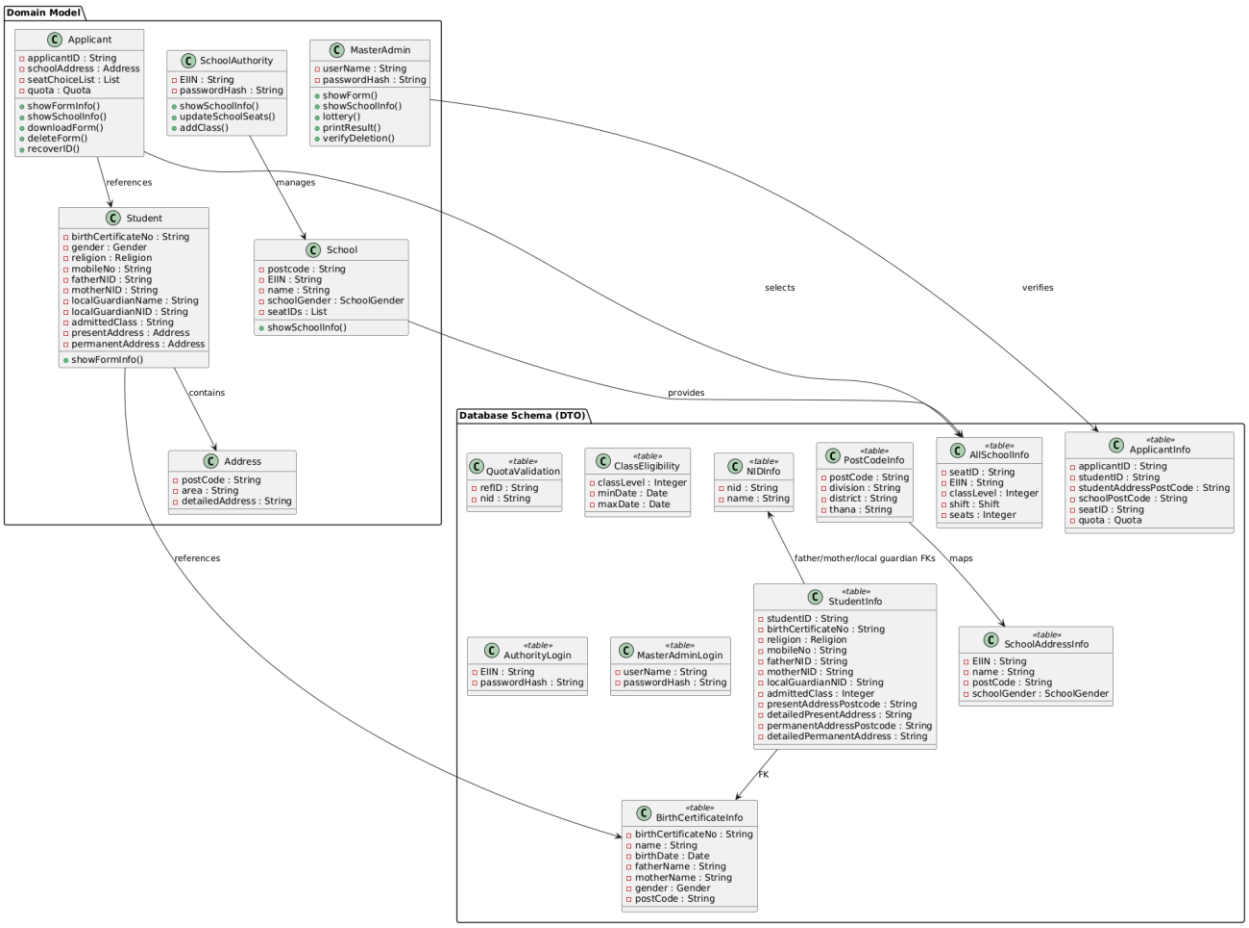
This School Admission Management System is a console-based application designed to simplify the school admission process. Applicants can enter their necessary information and apply to a list of preferred schools within a selected region. The Master Admin conducts the lottery process, and specific quotas—such as local school quota, freedom fighter quota, etc are applied during seat allocation. A dedicated School Admin panel allows each school to view and update its own information. Applicants also have the ability to delete, download, or recover their Application Form and ID.

Motivation

In Bangladesh, guardians experience several challenges during the school admission process. The current system often becomes time-consuming, requiring long queues and repeated visits to schools. Additionally, manipulative activities by middlemen frequently influence the process, creating unfair obstacles for genuine applicants.
Another major issue is biased selection, where deserving students may be overlooked due to unofficial influences. Furthermore, there is a significant lack of clear and accessible information regarding school requirements, seat availability, and admission procedures. This lack of transparency makes the process even more stressful for guardians.

Our project aims to recreate and improve the existing system by introducing a structured, transparent, and user-friendly School Admission Management System. Through automation and fair processing methods, the project intends to reduce manipulation, save time, and ensure equal opportunities for all applicants.

Class Diagram



Key features of the project

For applicants, the system ensures a smooth form-fill-up experience. Students can fetch their information by filling out birth certificate number, verify their mobile number through OTP, and even fill up their address automatically using only the postcode. The system will identify guardian information, checks document types like NID or passport, and validates all details—including class eligibility based on birth year. Applicants can preview their full form, re-edit if needed, apply for quotas with proper document references, [Automatically applied for local school quota and finally download a clean text-based form after confirmation. They can also recover their ID or delete a form through secure OTP verification. For seat information, users can browse available schools by district, view class-wise and shift-wise seat availability.
On the school side, school authorities can log in using their EIIN, update school status, manage seats, and configure classes and shifts.

Finally, the Master Admin holds full oversight. They can view any school or applicant form, authenticate removals, and perform the final lottery operation to ensure fairness and transparency.
In short, our system brings automation, validation, accuracy, and transparency into every step of the admission process, benefiting both applicants and administrators alike.

Tools and technology

Programming Language: The entire project is developed using Java.
IDEs: We used Visual Studio Code to write, test, and organize our code efficiently.
Version Control: For collaboration and tracking changes, we used Git.
Data Storage: Since databases were restricted for this project, we implemented a file-based storage system to save and manage all necessary information.

Proposed Timeline

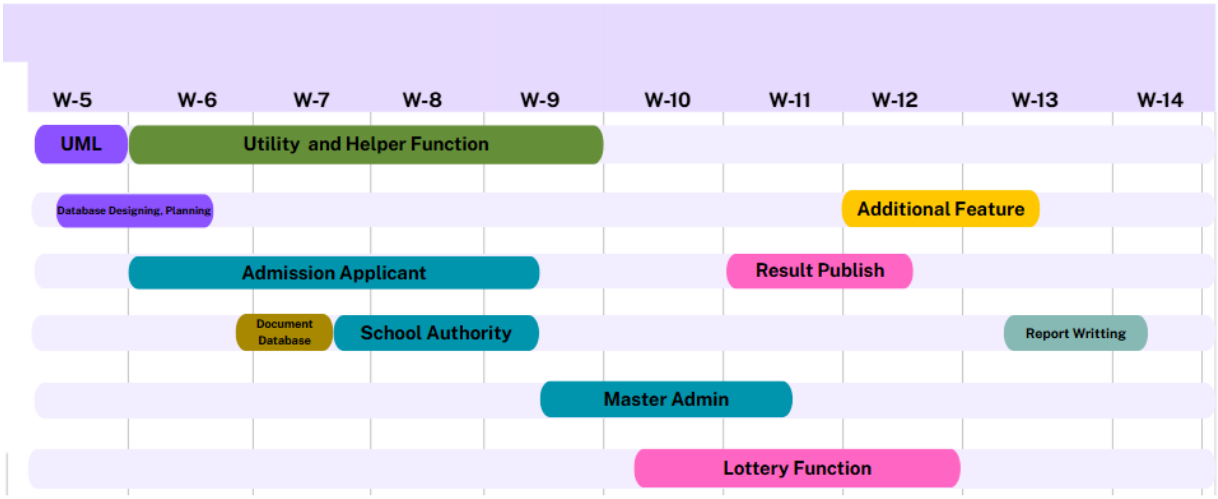
Week-5: Our team will focus on designing the complete UML diagram. By the middle of the week, we will also begin database design.

Week-6: The database design will be completed by mid-week. At the same time, we will start developing utility and helper functions, and also begin working on the Admission Applicants module. By the end of the week, we will inject initial information into the database.

Week-7 to Week-9: we will continue developing the Admission Applicants module, which will be completed by mid-week 9. Alongside this, we will work on the School Authority module. Both modules are expected to be finished before our Progression Presentation.

Mid-Week 9 to the end of Week 11: We will develop the Master Admin module and one of the core features, the Lottery function, which will take several weeks to complete.

Week-12: We will implement Result Publishing based on the lottery outcomes, add any additional features, and complete the final project report.


W-5
W-6
W-7
W-8
W-9
W-10
W-11
W-12
W-13
W-14
UML
Utility and Helper Function
Database Designing, Planning
Additional Feature
Admission Applicant
Result Publish
Document Database
School Authority
Report Writting
Master Admin
Lottery Function

Action taken based on the feedback

There are some feedback we got from the teacher so the feedback and the response of the feedback given below

Feedback 1:

There was two IDE VS Code and IntelliJ IDEA in our slide

Proposed solution: Teacher suggested use one IDE

Solution: We choose to use VS Code as our IDE

Feedback 2:

In our slide we write Mobile No. check by OTP in key feature of our web application

Proposed solution: Teacher said as it is a web-based solution it doesn't seems realistic.

Solution: So we will generate the OTP but sending mechanism will be abstract.

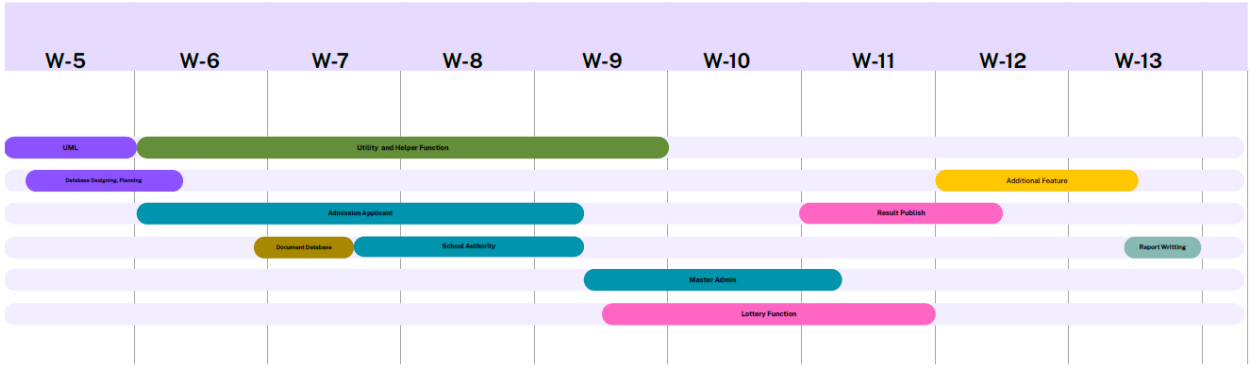
Feedback 3:

In Gantt chart font was small

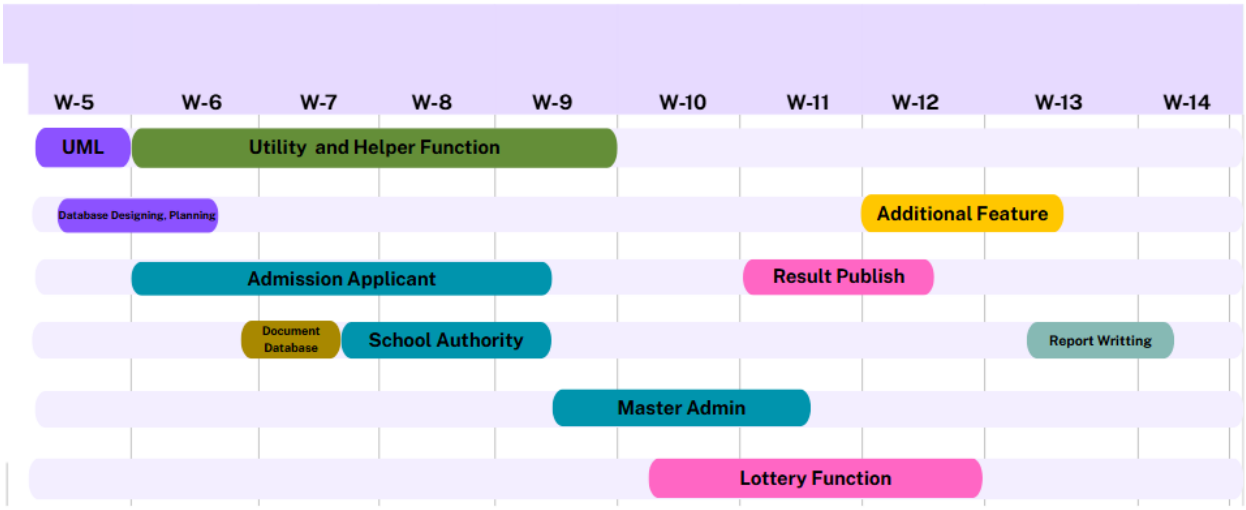
Proposed solution: Teacher suggested increase the size of font

Solution:

Before:



After:



There were some questions from teachers and we were able to convince them with our answers.